Fiscal Year:	FY 2026	**Task Last Updated:**	FY 05/20/2026
PI Name:	Robin, Adrien Ph.D.		
Project Title:	Effect of Deconditioning on Gravitational Dose-Response Curves for Cardiovascular and Ocular Variables in Men and Women		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	adrien.robin@tamu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	06-89-38-90-08 33
Organization Name:	Texas A&M Research Foundation		
PI Address 1:	701 Ross Street		
PI Address 2:			
PI Web Page:			
City:	College Station	**State:**	TX
Zip Code:	77843-0001	**Congressional District:**	10
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2023 TRISH-RFA-2301: Translational Research Institute for Space Health (TRISH) Postdoctoral Fellowships
Start Date:	12/01/2023	**End Date:**	09/30/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: New end date is 09/30/2025 per TRISH (Ed., 7/30/25).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Diaz Artiles, Ana Ph.D. (MENTOR: Texas A&M Engineering Experiment Station)		
Grant/Contract No.:	NNX16AO69A-P0803		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	POSTDOCTORAL FELLOWSHIP Exposure to weightlessness results in a permanent headward fluid shift, which might be related to spaceflight-induced neuro-ocular and functional changes, known as Spaceflight Associated Neuro-Ocular Syndrome (SANS). SANS has been diagnosed in both males and females, contrary to initial reports suggesting a male-only condition. While the exact etiology of SANS is currently unknown, chronic fluid redistribution affecting intravascular, interstitial, and cerebrospinal fluids and pressures is widely hypothesized to be a contributing factor. Additionally, recently demonstrated stagnant and retrograde blood flow and venous thrombosis in the jugular vein during spaceflight could also be associated with sustained headward blood and tissue fluid shift. Upcoming long-duration missions to the Moon and Mars will require months or even years in altered gravity conditions. Thus, there is a need to develop novel countermeasures, ensuring astronauts are healthy and fully operational on return to a gravitational environment. Specifically, countermeasures focused on reintroducing the hydrostatic gradients and counteracting the fluid shift are particularly relevant – for example, lower body negative pressure (LBNP) or centrifugation. But for those to be applied successfully, we need to better understand the overall physiological responses to particular “gravity dose”. Our previous work has focused on gravitational dose-response from healthy individuals. Assuming comparable gravitational dose responses in "deconditioned" subjects is most likely inappropriate; therefore, it requires further investigation to better tailor countermeasure protocols. In this research effort, we propose to generate acute gravitational dose-response curves of fluid shift, cardiovascular, and ocular variables on “deconditioned” subjects using a tilt paradigm. Healthy subjects (16 males and 16 females) will experience two different “deconditioning” interventions: 1) 24h head down bed-rest (HDBR) and 2) drug-induced rapid hypovolemia (i.e., furosemide) in a crossover design. Before (i.e., “healthy” conditions), and after the “deconditioning” interventions, subjects will be exposed to multiple tilt angles (from 45° head-down tilt/HDT to 45° head-up tilt/HUT in supine position) to generate gravitational dose responses to tilt. Measures include calf volume, segmental hydration state, blood volume, continuous cardiovascular hemodynamic variables, superficial venous network, skin microcirculation, muscle and cerebral blood flow, characteristics of the carotid and jugular veins (cross-sectional area and jugular vein pressure), optic nerve sheath diameter, intra-ocular pressure, and ophthalmic artery perfusion. We hypothesized that dose-response curves after deconditioning will differ from the “healthy” dose-response curves for most variables. We will also perform the experiments in both males and females to further characterize sex differences in these responses. The specific objectives of this proposal are summarized below: - Specific Aim 1: To generate gravitational dose-response relationships of fluid shift, cardiovascular, and ocular parameters as a function of gravity level in deconditioned individuals (24h HDBR or drug-induced hypovolemia) using a tilt paradigm. - Specific Aim 2: To characterize differences between deconditioning interventions: 24 HDBR vs drug-induced hypovolemia. - Specific Aim 3: To compare sex differences in dose-responses between males and females. Our study will generate a comprehensive conceptual framework for fluid-shift response prediction. This framework will constitute an important terrestrial model to reference spaceflight-induced changes. It will contribute to the assessment of the headward fluid shift, the pathogenesis of SANS and spaceflight venous thromboembolism events, and help in development of countermeasures and in-flight prescriptions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our study will generate a comprehensive conceptual framework for fluid-shift response prediction. This framework will constitute an important terrestrial model to reference spaceflight-induced changes, and it will contribute to the assessment of the headward fluid shift, the pathogenesis of SANS and spaceflight venous thromboembolism events, as well as inform the development of countermeasures and in-flight prescriptions.
Task Progress:	We are still actively recruiting participants for the bedrest campaign, to complete Aim 1 with bedrest-related dose-response curves, and to proceed with Aims 2 and 3 with a comparison between the two deconditioning interventions (i.e., diuretic vs. bedrest) and between males and females to assess sex differences. Thus far, four participants have been tested, we expect to recruit and test more participants in the next few months.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
Articles in Peer-reviewed Journals	Robin A, Zaman SY, Reinarz C, Lawson K, McKnight J, Navasiolava N, Custaud M, Haddad L, Zawieja D, Diaz-Artiles A. "Ocular and jugular gravitational dose-response curves during hypovolemia: Preliminary results." Physiology. 2025 May 12;40(S1):0116. https://journals.physiology.org/doi/abs/10.1152/physiol.2025.40.S1.0116 , May-2025
Articles in Peer-reviewed Journals	Robin A, Reinarz CD, Pentinat-Llurba H, Zaman SY, Rasmussen CJ, McKnight JR, Haddad L, Zawieja DC, Diaz-Artiles A. "Dose-dependent effects of graded altered gravity during hypovolaemia on central haemodynamics and cardiovascular autonomic regulation." Exp Physiol. 2026 Jun 19. Online ahead of print. https://doi.org/10.1113/EP093840 ; PMID: 42321023 , Jun-2026
Awards	Robin A. "Walter B. Cannon Lecture Abstracts of Distinction, April 2025." Apr-2025